

Facial Recognition Bias Audit Student Handout

TOK Cognitive Science Activity Bank • Knowledge and Technology • 35-45 min

Category	Knowledge and Technology	Time	35-45 min
Difficulty	Medium	ID	facial-recognition-bias-audit

Big idea: Automated recognition is a knowledge claim whose reliability can vary across contexts and groups.

Student-Facing Prompt

Inspect Facial Recognition Bias Audit Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Facial Recognition Bias Audit Stimulus A	Student-facing stimulus 1: Audit row: hallway camera, 94% overall accuracy, but higher false positives in low light. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Facial Recognition Bias Audit Stimulus B	Student-facing stimulus 2: Stakeholder card: student wrongly flagged, administrator managing safety, developer defending the model. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Facial Recognition Bias Audit Reveal / Twist	Student-facing stimulus 3: Stakeholder card: student wrongly flagged, administrator managing safety, developer defending the model. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Automated recognition is a knowledge claim whose reliability can vary across contexts and groups.

Actual Lesson Questions

#	Question for students
1	After inspecting Facial Recognition Bias Audit Stimulus A, what is your first knowledge claim?
2	Which exact detail from Facial Recognition Bias Audit Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	When is a technological classification reliable enough to act on?

Ready-to-Use Examples

- Audit row: hallway camera, 94% overall accuracy, but higher false positives in low light.
- Stakeholder card: student wrongly flagged, administrator managing safety, developer defending the model.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Activity Steps

1. Introduce a fictional school security tool and its claimed accuracy.
2. Groups inspect audit data broken down by context and subgroup.
3. Students identify where the headline accuracy hides reliability differences.
4. Role groups decide whether to deploy, revise, restrict, or reject the tool.
5. Debrief how technical evidence and ethical responsibility interact.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: When is a technological classification reliable enough to act on?
- Can a system be accurate overall and still unjustified in practice?
- What evidence is needed before automated knowledge claims affect people?

Knowledge Questions

- When is a technological classification reliable enough to act on?
- How should knowers weigh overall accuracy against uneven error?
- Who is accountable for knowledge produced by automated systems?

Transfer Example

For example, run this activity with mock recognition results with different error rates across groups. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...